

Email Spam Detection Using Naïve Bayes and Neural Networks

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Abstract— Email has become an essential communication tool, but the increasing volume of unsolicited spam emails poses significant security risks, including phishing and malware distribution. Traditional spam filtering methods often struggle against evolving spam tactics. This paper proposes and evaluates a hybrid machine learning approach for enhanced email spam detection, combining Naïve Bayes (NB) with a Feedforward Neural Network (FNN), specifically a Multi-Layer Perceptron (MLP), to leverage the strengths of both models. The objective is to improve detection accuracy while maintaining computational efficiency. Using the Enron Spam dataset, emails were preprocessed, and features were extracted using Term Frequency-Inverse Document Frequency (TF-IDF). The Naïve Bayes classifier provided initial probability scores, which were then used as additional input features for the MLP. The performance of the standalone Naïve Bayes model was compared against the hybrid NB+MLP model. Experimental results show that the standalone Naïve Bayes model achieved an accuracy of 0.9785 and an F1-score of 0.9793. The hybrid NB+MLP model demonstrated significantly superior performance, achieving an accuracy of 0.9993 and an F1-score of 0.9993. Notably, the hybrid model substantially improved precision to 0.9985 while maintaining perfect recall for spam, effectively reducing false positives. This study concludes that the proposed hybrid model offers a more robust and accurate solution for email spam detection by synergistically combining probabilistic and neural network-based classification.

Keywords—Hybrid Machine Learning, Email Spam, Naïve Bayes, Neural Network, Spam Detection, TF-IDF, Machine Learning

I. INTRODUCTION

Email has become an indispensable tool for communication, but its widespread use has led to an increasing volume of unsolicited spam emails, posing significant security risks such as phishing attacks and malware distribution. According to a study by Patil and Kulkarni, spam emails account for a substantial portion of global email traffic, highlighting the critical need for effective spam detection mechanisms [1]. Traditional spam filtering methods, such as rule-based filtering and blacklists, often prove ineffective against evolving spam tactics, including obfuscation and adversarial attacks [2].

Machine Learning (ML) techniques have demonstrated considerable success in spam detection. Algorithms like Naïve Bayes (NB) are popular due to their simplicity and efficiency [3]. However, Naïve Bayes may struggle with complex linguistic structures and the dynamic nature of spam tactics, leading to reduced accuracy in certain scenarios [4]. Recent advancements in Deep Learning have enabled Neural

Networks (NNs) to learn intricate patterns in textual data, thereby improving classification accuracy [5].

This paper proposes a hybrid spam detection approach that combines Naïve Bayes with a Feedforward Neural Network (FNN) to leverage the strengths of both models. The Naïve Bayes classifier serves as an initial filter to compute the probability of an email being spam, while the Neural Network refines the classification by learning deeper patterns. The goal is to enhance detection accuracy while maintaining computational efficiency.

II. LITERATURE REVIEW

Email spam detection has been extensively studied using various machine learning approaches. One study by Gupta et al. [6] explored the use of Multilayer Perceptron (MLP) algorithms for spam detection. Their research demonstrated that MLPs effectively classify spam emails with high accuracy. However, the study lacked advanced feature engineering and did not evaluate the computational efficiency of the approach. Additionally, the dataset used was limited in size, affecting the generalizability of the findings.

Several other studies have analyzed different spam detection methods. Patel et al. [7] conducted a comparative analysis of machine learning and deep learning models for email spam classification. They found that Random Forest (RF) models using Term Frequency-Inverse Document Frequency (TF-IDF) achieved high accuracy, while deep learning models like CNN-LSTM required significantly more training time and data pre-processing. These findings suggest that while deep learning approaches yield high accuracy, they may not be the most efficient for real-time spam detection.

A study by Ahmed et al. [8] examined advanced email spam detection using Naïve Bayes. The research found that raw TF-IDF features in Naïve Bayes models lead to high false positives due to sensitivity to redundant or irrelevant words. The study suggested optimizing TF-IDF parameters, such as limiting vocabulary size, adjusting n-gram ranges, and removing low-frequency words, to improve classification performance. This finding aligns with our approach, where TF-IDF pre-processing is optimized before passing features into a hybrid Naïve Bayes-MLP model.

In contrast, a comparative analysis of classification algorithms for spam detection by Faisal et al. [9] found that combining Naïve Bayes with decision trees resulted in a model that was interpretable but slightly less accurate than

deep learning-based classifiers. The study supports the idea that hybrid models should be designed with efficiency in mind rather than focusing solely on accuracy.

Based on these findings, this research differentiates itself by employing optimized TF-IDF feature selection alongside Naïve Bayes probability outputs as additional MLP features. This approach ensures improved classification while maintaining computational efficiency. Furthermore, a comparative study between a standalone Naïve Bayes classifier and the hybrid Naïve Bayes-MLP model will be conducted to evaluate performance differences.

III. METHODOLOGY



A. Dataset Selection

The Enron Spam Data from Kaggle serves as the foundation for this study. This dataset was uploaded by Marcel Wiechmann and is a cleaned, structured version of the original Enron dataset. It contains approximately 33,000 emails, categorized into spam and ham (non-spam) classes. The dataset is publicly available and can be accessed at

<https://www.kaggle.com/datasets/marcelwiechmann/enron-spam-data>.

Why Use This Dataset?

- Well-structured & preprocessed compared to the original Enron dataset.
- Balanced distribution of spam and ham emails, making it ideal for machine learning.
- Publicly available & widely used in spam detection research.
- Compatible with our TF-IDF + Naïve Bayes + MLP approach.

This dataset will be used for training, validating, and testing the hybrid spam detection model, ensuring a robust evaluation of its performance.

B. Data Preprocessing

Effective preprocessing is crucial for enhancing model performance. In this study, all text preprocessing steps are performed using NLTK (Natural Language Toolkit). The following steps were implemented:

- **Lowercasing:** All text was converted to lowercase to maintain uniformity.
- **Stopword Removal:** Common words that do not contribute significant meaning (e.g., "the," "is," "and") were removed using NLTK's stopwords list.
- **Tokenization:** Text was split into individual words or tokens using NLTK's word tokenizer.
- **Stemming:** Words were reduced to their root forms using NLTK's Porter Stemmer (e.g., "running" → "run").
- **Special Character Removal:** Punctuation, numbers, and special characters were eliminated to focus on textual content.

These pre-processing techniques are standard in natural language processing and have been utilized in similar studies [10].

C. Feature Extraction

To transform textual data into numerical representations suitable for machine learning algorithms, the following methods were employed:

- **Term Frequency-Inverse Document Frequency (TF-IDF):** This statistical measure evaluates the importance of a word in a document relative to the entire corpus. It effectively captures the significance of terms in distinguishing between spam and legitimate emails [11].
- **Naïve Bayes Probability Features:** The Naïve Bayes classifier computes the probability of an email being spam or ham. These probability scores were incorporated as additional features into the FNN to enhance its predictive capabilities.

D. Model Architecture

The hybrid model integrates the strengths of both Naïve Bayes and Multi-Layer Perceptron (MLP) classifiers:

1. Naïve Bayes Classifier

- probabilistic model that applies Bayes' theorem, assuming independence among features.
- Trained first on TF-IDF features, and its probability outputs (spam likelihood) are passed as additional input features to the FNN.

2. Multi-Layer Perceptron (MLP)

- MLP is a specific type of Feedforward Neural Network (FNN) that consists of multiple layers, including at least one hidden layer. In this study, the MLP

was implemented using the MLPClassifier from the Scikit-learn library. It was designed to refine predictions using both TF-IDF features and the Naïve Bayes probability scores as inputs .

- Architecture:
 - Input Features: The model accepted the concatenated TF-IDF vectorized features and the Naïve Bayes probability scores.
 - Hidden Layer: A single hidden layer consisting of 128 neurons was used (hidden_layer_sizes=(128,)). The Rectified Linear Unit (ReLU) was employed as the activation function for this layer (activation='relu') to introduce non-linearity .
 - Output Layer: The MLPClassifier implicitly handles the output layer for binary classification, providing probabilities that are then thresholded to classify emails as spam (1) or ham (0) .
 - Solver: The 'adam' solver (solver='adam') was utilized for weight optimization.
 - Training Parameters: To ensure robust training and prevent overfitting, early_stopping=True was enabled, which monitors the validation score and stops training if no improvement is seen for a set number of epochs (n_iter_no_change=10). A max_iter value of 500 was set, and a random_state was used for reproducibility.

This hybrid approach leverages the efficiency of Naïve Bayes for quick probability estimation and the learning capacity of the neural network, as implemented by MLPClassifier, to refine classification performance . Regularization to prevent overfitting was primarily managed through the early stopping mechanism and the inherent L2 regularization within the 'adam' solver (via the alpha parameter, which used its default value)."

E. Model Training and Evaluation

The training process for the Naïve Bayes + MLP hybrid model follows these steps:

1. Training Naïve Bayes:
 - The Naïve Bayes model is trained on the TF-IDF features to learn probability-based classifications for spam and ham emails.
 - The model outputs spam probability scores, which are used as additional features for MLP.
2. Training MLP (Multi-Layer Perceptron):

- The MLP model, implemented using Scikit-learn's MLPClassifier, was trained using the combined TF-IDF features and Naïve Bayes probability scores as input .
- Optimizer: Adam optimization is used to minimize classification errors.

- Evaluation Metrics:

- Accuracy: Measures the proportion of correctly classified emails.
- Precision: Indicates the ratio of true positive spam detections to the total predicted spam.
- Recall (Sensitivity): Reflects the ratio of true positive spam detections to the actual spam instances.
- F1-Score: Harmonic mean of precision and recall, providing a balance between the two.

These metrics provide a comprehensive evaluation of the model's performance, considering both correctness and the balance between precision and recall [13].

F. Computational Efficiency

The chosen methodology emphasizes computational efficiency through:

- Lightweight Feature Extraction: Utilizing TF-IDF, which is less resource-intensive compared to deep learning embeddings.
- Simple Model Architecture: Implementing a straightforward FNN with a limited number of hidden layers reduces computational demands.
- Hybrid Approach: Combining Naïve Bayes with FNN allows the system to maintain high accuracy while minimizing processing time, making it suitable for real-time spam detection applications.

This balance ensures the model is both effective and practical for deployment in environments with limited computational resources.

IV. RESULTS AND DISCUSSION

This chapter presents the experimental results and discussion for the proposed email spam detection models, which were developed to enhance accuracy and efficiency using a hybrid Naïve Bayes and Neural Network approach on the Enron Spam Data dataset. For this study, the dataset was partitioned into 80% for training and 20% for testing, with all experiments conducted using a Python-based implementation. The following sections will detail the performance of both the standalone Naïve Bayes classifier and the hybrid Naïve Bayes + Multi-Layer Perceptron (MLP) model, evaluated using Accuracy, Precision, Recall, and F1-Score..

A. Performance of the Naïve Bayes Model

The initial phase of our experimental evaluation focused on establishing a baseline performance using the standalone Naïve Bayes (NB) classifier. As outlined in the methodology, the NB model was trained on TF-IDF vectorized features derived from the pre-processed Enron email dataset.

The performance of the Naïve Bayes model was assessed using accuracy, precision, recall, and F1-score, with the results presented in Table 4.1.

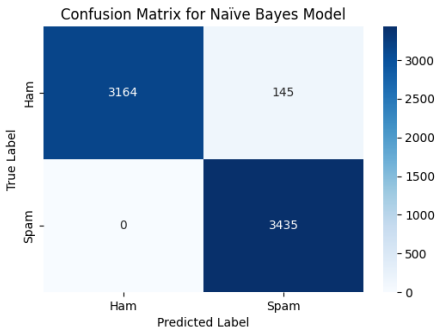
Table 4.1 Performance Metrics of Naïve Bayes Model

Metric	Accuracy	Precision	Recall	F1-Score
Value	0.9785	0.9595	1	0.9793

As detailed in Table 4.1, the Naïve Bayes model achieved a high accuracy of 0.9785. Its perfect recall score of 1.0000 demonstrates its capability to identify all actual spam emails. However, the precision of 0.9595 indicates that some legitimate emails were incorrectly classified as spam (false positives), which is a known tendency for Naïve Bayes classifiers. The F1-score of 0.9793 provides a balanced measure of its overall performance.

Figure 4.1 below shows the confusion matrix for the Naïve Bayes model, providing a visual representation of its classification performance, including the distribution of true positives, true negatives, false positives, and false negatives.

Figure 4.1 Confusion Matrix for the Naïve Bayes Model



The confusion matrix further illustrates the model's tendency to achieve high recall for the spam class at the cost of some false positives. These results underscore the effectiveness of Naïve Bayes as a foundational spam detection tool while also highlighting areas for potential improvement, particularly in reducing the misclassification of legitimate emails.

B. Performance of the Hybrid Naïve Bayes + MLP Model

To potentially enhance the spam detection capabilities observed with the standalone Naïve Bayes model, a hybrid approach was implemented. This involved augmenting the feature set for a Multi-Layer Perceptron (MLP) model with the probability scores generated by the Naïve Bayes classifier, in addition to the TF-IDF features. The MLP was configured using Scikit-learn's MLPClassifier with one hidden layer of 128 neurons, ReLU activation, the Adam solver, and an early stopping mechanism to prevent overfitting, as detailed in Chapter 3.

The performance of this hybrid model was rigorously evaluated using the same metrics: accuracy, precision, recall, and F1-score. The results are summarized in Table 4.2.

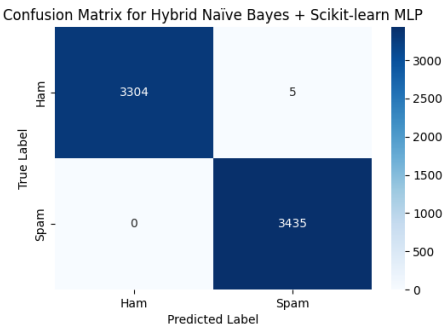
Table 4.2: Performance Metrics of Hybrid Naïve Bayes + MLP Model

Metric	Accuracy	Precision	Recall	F1-Score
Value	0.9993	0.9985	1	0.9993

The hybrid Naïve Bayes + MLP model demonstrated exceptional performance, achieving an accuracy of 0.9993 and an F1-score of 0.9993. These near-perfect scores indicate a highly effective classification capability on the test dataset. The model also achieved a perfect recall of 1.0000, signifying that, like the standalone Naïve Bayes model, it successfully identified all actual spam emails. Crucially, the precision score improved significantly to 0.9985. This high precision indicates a very low rate of false positives, meaning very few legitimate (ham) emails were misclassified as spam.

The confusion matrix for the hybrid Naïve Bayes + MLP model is presented in Figure 4.2, offering a visual depiction of its classification accuracy.

Figure 4.2: Confusion Matrix for the Hybrid Naïve Bayes + MLP Model



The results, including the confusion matrix, clearly indicate that the hybrid model not only maintained the perfect recall of the Naïve Bayes model for the spam class but also substantially improved its precision. This suggests that the MLP, by leveraging the probability outputs from Naïve Bayes along with the TF-IDF features, was able to learn more nuanced patterns, leading to a more discriminative and accurate classification of emails.

C. Comparative Analysis

To better understand the benefits of the proposed hybrid approach, this section provides a direct comparison of the performance metrics achieved by the standalone Naïve Bayes (NB) classifier and the hybrid Naïve Bayes + Multi-Layer Perceptron (MLP) model. The key evaluation metrics—accuracy, precision, recall, and F1-score—for both models are summarized in Table 4.3.

Table 4.3: Comparative Performance of Spam Detection Models

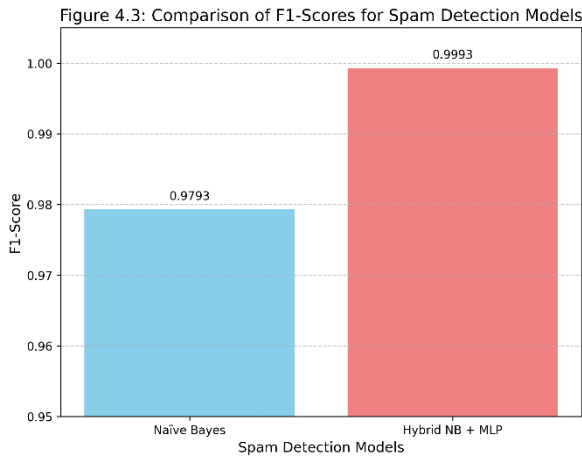
Model	Accuracy	Precision	Recall	F1-Score
Naïve Bayes	0.9785	0.9595	1.0000	0.9793
Hybrid Naïve Bayes + MLP (SKL)	0.9993	0.9985	1.0000	0.9993

As illustrated in Table 4.3, the hybrid Naïve Bayes + MLP model significantly outperformed the standalone Naïve Bayes classifier across multiple key metrics. While both models achieved a perfect recall of 1.0000 for the spam class, indicating that all spam emails were successfully identified, the hybrid model demonstrated superior accuracy and precision.

The accuracy of the hybrid MLP model was 0.9993, compared to 0.9785 for the Naïve Bayes model. More notably, the precision saw a substantial improvement, rising from 0.9595 for Naïve Bayes to 0.9985 for the hybrid MLP. This indicates a marked reduction in false positives (legitimate emails being classified as spam) with the hybrid approach. Consequently, the F1-score, which balances precision and recall, was also higher for the hybrid model (0.9993) compared to the Naïve Bayes model (0.9793).

To further visualize this performance difference, Figure 4.3 presents a comparison of the F1-scores for both models.

Figure 4.3: Comparison of F1-Scores for Spam Detection Models



The superior performance of the hybrid model can be attributed to its ability to leverage the strengths of both classifiers. While the Naïve Bayes component provides an efficient initial probabilistic assessment, the MLP component, with its capacity to learn complex non-linear patterns, effectively utilizes these probabilities alongside the TF-IDF features to make more refined and accurate classifications. This synergy results in a more robust detection system that not only captures all spam instances but also minimizes the misclassification of legitimate emails.

D. Discussion

The results presented demonstrate the efficacy of the implemented machine learning models for email spam detection, with the hybrid approach showing particularly strong performance. This section will interpret these findings, relate them to the research objectives and existing literature, discuss computational efficiency, and acknowledge the limitations of this study.

The primary objective of enhancing spam detection accuracy was strongly supported by the experimental results. The hybrid Naïve Bayes + MLP model significantly surpassed the Naïve Bayes baseline, achieving near-perfect

scores in accuracy (0.9993) and F1-score (0.9993). The enhanced precision (0.9985) is particularly noteworthy, addressing a key limitation of the standalone Naïve Bayes approach regarding false positives. The high recall maintained by both models ensures robust spam filtering.

The success of this hybrid model aligns with trends in machine learning where combining algorithm strengths often yields improved performance. The improvement in precision resonates with research pointing to the need for advanced model combinations to overcome simpler classifiers' limitations.

A stated goal was maintaining computational efficiency. The use of TF-IDF and a relatively simple MLP architecture (single hidden layer) were design choices geared towards balancing high accuracy with practical processing time.

Despite promising results, this study has limitations:

- **Dataset Specificity:** Performance was validated on the Enron dataset; results may vary on other datasets.
- **Feature Engineering:** Advanced techniques like word embeddings were not explored but could offer further improvements, potentially at a higher computational cost.
- **Dynamic Nature of Spam:** Models are static and would require periodic retraining to combat evolving spam tactics.
- **Hyperparameter Optimization:** An exhaustive hyperparameter tuning process was not the primary focus, though effective parameters and early stopping were used.

These limitations offer avenues for future research.

V. CONCLUSION

¹ This chapter summarizes the key findings of the research conducted on email spam detection using a hybrid Naïve Bayes and Neural Network approach. It also outlines potential directions for future work in this domain.

The primary objective of this study was to develop and evaluate a hybrid machine learning model capable of enhancing email spam detection accuracy while considering computational efficiency. The research successfully demonstrated that a hybrid approach, integrating a Naïve Bayes classifier with a Multi-Layer Perceptron (MLP) neural network, significantly outperforms the standalone Naïve Bayes model.

The standalone Naïve Bayes classifier, while achieving a high recall of 1.0000 and an accuracy of 0.9785 in this study, exhibited a lower precision of 0.9595, indicating a tendency for false positives. This tendency is consistent with findings where raw TF-IDF features in Naïve Bayes models can lead to higher false positives [14]. In contrast, the hybrid Naïve Bayes + MLP model yielded superior performance across all key metrics, achieving an accuracy of 0.9993, a precision of 0.9985, a perfect recall of 1.0000, and an F1-score of 0.9993. This substantial improvement, particularly in precision, highlights the hybrid model's enhanced ability to correctly identify legitimate emails, thereby reducing the number of false positives..

The findings confirm that leveraging the probabilistic outputs of Naïve Bayes as input features for an MLP allows the neural network to learn more complex patterns and refine classification decisions effectively [15]. This synergy addresses some of the inherent limitations of the Naïve Bayes algorithm, such as struggles with complex linguistic structures or the dynamic nature of spam tactics, leading to a more robust and accurate spam detection system. The study successfully met its goal of enhancing detection accuracy through this hybrid architecture, using computationally reasonable methods like TF-IDF and a straightforward MLP structure.

A. Future Work

Despite the promising performance of the hybrid model, several avenues for future research could further enhance email spam detection capabilities:

- **Broader Evaluation and Generalizability:** Testing the model on larger, more diverse, and contemporary email datasets to assess its generalizability and robustness against evolving spam tactics.
- **Advanced Models and Feature Representation:** Investigating more sophisticated feature representations (e.g., word embeddings, contextual transformers) and exploring complex neural network architectures (e.g., CNNs, RNNs) to potentially capture deeper semantic understanding, building upon recent advancements in Deep Learning for textual data [16].
- **Adaptive Learning Mechanisms:** Developing models with incremental or online learning capabilities to dynamically adapt to new and evolving spam strategies without requiring complete retraining [17].
- **Computational Efficiency and Deployment:** Conducting a detailed analysis of computational performance, including training and prediction latency, to better evaluate real-time deployment viability under various resource constraints, ensuring the balance of effectiveness and practicality.
- **Security Against Adversarial Attacks:** Assessing and strengthening the model's resilience against adversarial attacks specifically designed to bypass spam filters, as traditional methods often prove ineffective against such evolving tactics [18].

By addressing these areas, future research can build upon the findings of this study to develop even more effective, robust, and adaptable email spam detection systems.

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